Detecting task events in fMRI time series based on visibility graphs

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INTRODUCTION

- Identification of extreme events (EV) in time-series (TS) is a critical aspect in decoding **task-modulated brain activity** in fMRI.
- Task fMRI applicability has been often limited to **linear modeling** of the TS, usually by applying a convolution with an **HRF** over the TS and estimating the fit of an expected versus the actual TS shape under task events
- This type of modeling **assumes uniformity** of the TS shape across contexts, whereas the stimulus-induced BOLD behavior demonstrates considerable variability [1]

VISIBILITY GRAPHS (VG)

- **Visibility graphs** (VG) can be used to **convert TS into graphs** [2]
- Allow using tools of complex systems to analyze TS
- They remain immune to participant-specific differences in signal amplitude and preserve the structure despite the existence of linear trends in the TS [2]
- The TS is conceptualized as a **network of time points** (in fMRI each TR would be a node of the temporal network)
- A **connection** is attributed geometrically, based on whether or not a node could "see" another node
- TPs that have **higher intensity** (such as signal stimulus-evoked spikes in the event-related designs) "sit" higher in the TS, thus being able to have a **higher visibility** over other time points (TP) in the TS
- They therefore have a **higher degree** (more edges) in the TS network

OBJECTIVES

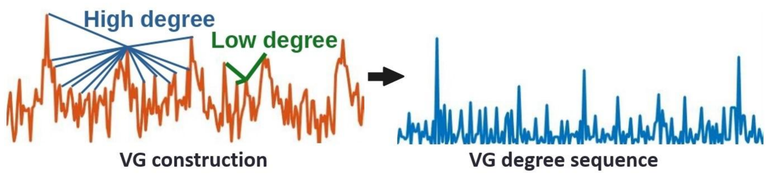
- The aim of the current analysis is to test whether VG could successfully **identify stimulus-evoked TS changes** in event-related design:
  - Using simulated data
  - Using real event-related fMRI data from the BOLD5000 open dataset (picture presentation task) [3]

METHODS

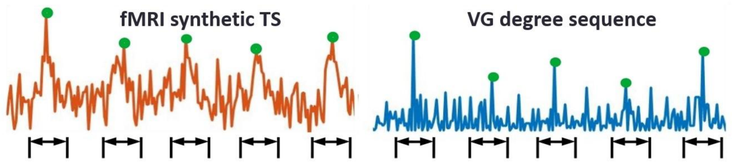
- **Synthetic data** was generated to resemble an event-related fMRI design by simulating repetitive occurrences of HRF and gradually adding Gaussian noise (TS length = 194 TPs, 37 onsets 5 seconds apart)

METHODS

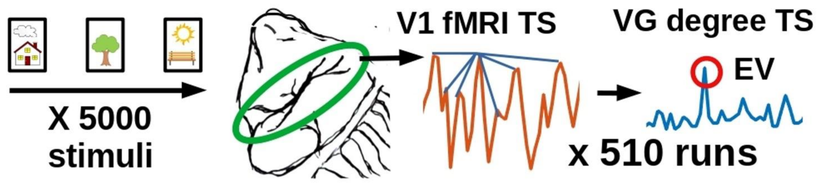
- **VG degree** sequence has been obtained by building the graphs over the whole synthetic TS, resulting in a new TS of the VG degree adjacent to each TP



- The **accuracy** in the two TS (VG degree vs. raw signal) has been estimated by counting the occurrence of signal spikes around HRF peaks versus the number of signal spikes that occur in arbitrary locations



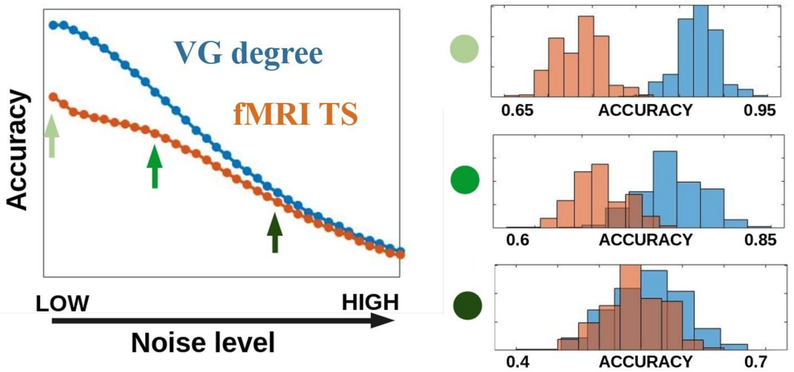
- This process has been repeated after adding **different amounts of Gaussian noise**, in order to understand the impact of noisy spikes in the accuracy
- **BOLD 5000** was used for replicating the results on real data, by extracting signal in the occipital cortex (V1) during the presentation of pictorial stimuli
- VG degree was constructed in the same way, over scanning runs and sessions coming from the 4 participants (510 TS in total)
- VG versus raw signal accuracy was obtained by counting the occurrence of VG degree and TS peaks around 6 s (3 +/- 2 TR) post-stimulus onset



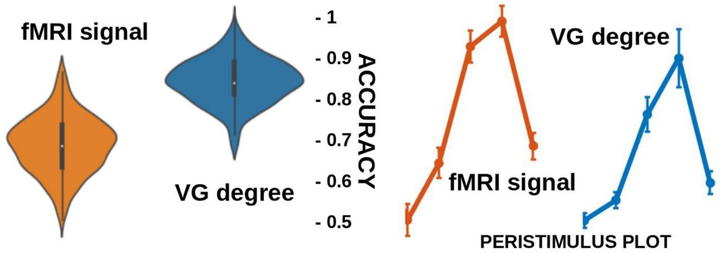
RESULTS

- The *event-related* synthetic fMRI data showed significantly **better accuracy for VG degree** as compared to raw signal
- This effect is **dependent on the amount of noise** added to the TS as the difference between VG and synthetic TS tends to decrease with increasing noise, likely due to the occurrence of spurious spikes induced by noise

RESULTS



- The replication on BOLD 5,000 dataset confirms significantly **higher accuracy of VG degree** (M = 0.84 +/- 0.05) in identifying EV ($p < 0.001$, two sided permutation test with 10,000 iterations) compared to fMRI signal (M= 0.7 +/- 0.07)
- **Peristimulus plots** exhibit a peak around 6 s post-stimulus presentation for both TS and VG degree.



CONCLUSIONS

- VG degree can accurately identify stimulus evoked EV in TS
- VG allow reducing the number of assumptions about the shape of the EV
- Instead of relying on the fit of an expected shape, the node degree may be a more intuitive measure of stimulus-evoked changes in the signal
- Having an accuracy measure could provide an informative and more direct estimate of the effect, instead a simplistic on/off activation approach
- The accuracy measure could be useful also to estimate the amount of noise in the data (for example when testing different denoising pipelines)
- Unlike modeling based on the GLM, VG degree allows approximating the location of the post-stimulus peak response

REFERENCES

- [1] Handwerker, D. A., Gonzalez-Castillo, J., D'esposito, M., & Bandettini, P. A. (2012), 'The continuing challenge of understanding and modeling hemodynamic variation in fMRI', Neuroimage, 62(2), 1017-1023.
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- [3] Chang, N., Pyles, J. A., Marcus, A., Gupta, A., Tarr, M. J., & Aminoff, E. M. (2019). BOLD5000, a public fMRI dataset while viewing 5000 visual images. *Scientific data*, *6*(1), 1-18.